

CM1605 Web Technology

CSS – Cascading Style Sheets

Week 3 | Janani Harischandra

Learning Outcomes

- Covers LO2 for Module Descriptor
- On completion of this lecture, students are expected to be able to:
 - Understand the concepts of cascading style sheets, CSS syntax,, CSS Box Model, Pseudo Classes & Elements, CSS selectors, CSS Positioning
 - Describe how CSS files work and give examples of CSS formatting tags.
 - Understand and apply different ways of using CSS in hypertext documents.

Content

- Introduction to CSS
- CSS Syntax : Property & Values
- CSS Formatting Tags
- How to use CSS
- CSS Selectors
- CSS Pseudo Classes & Elements
- CSS Box Model
- How CSS files Cascade
- Positioning with CSS
- CSS files validation & CSS3

CSS – Cascading Style Sheets

- CSS allows you to create rules that specify how the content of an element should appear.
- CSS is used to define styles for your web pages, including the design, layout and variations in display for different devices and screen sizes.
- The separation of HTML from CSS makes
 - Easier to maintain sites
 - Share style sheets across pages
 - Tailor pages to different environments
 - Reliable browser support
 - Layout controls
- CSS **saves a lot of work**. It can control the layout of multiple web pages all at once.

Power of CSS



CSS Zen Dragen
by Matthew Buchanan



Shaolin Yokobue
by Javier Cabrera



By the Pier
by Peter OngKelmscott



Organica Creativa
by Eduardo Cesario

All of these designs use the exact same HTML source document. but the design is changed using exclusively CSS (used with permission of CSS Zen Garden and the individual designers).

CSS Syntax

- CSS works by associating rules with HTML elements.
- These rules govern how the content of specified elements should be displayed.
- A CSS rule contains two parts: a **selector** and a **declaration**.
- **Selectors**
 - Indicate which element the rule applies to.
 - The same rule can apply to more than one element if you separate the element names with commas.
- **Declarations**
 - Indicate how the elements referred to in the selector should be styled.
 - Declarations are split into two parts (a property and a value), and are separated by a colon.

CSS Syntax



CSS Property and Values

- **Properties** indicate the aspects of the element you want to change.
- **Values** specify the settings you want to use for the chosen properties.
- Eg:
 - text-align : center;
 - color : black;
 - font-family : arial;
 - font-size : 12px;

CSS Formatting Categories

- Color – Background, Font
- Position
- Size
- Alignment
- Margins
- Text
- Borders

How to use CSS

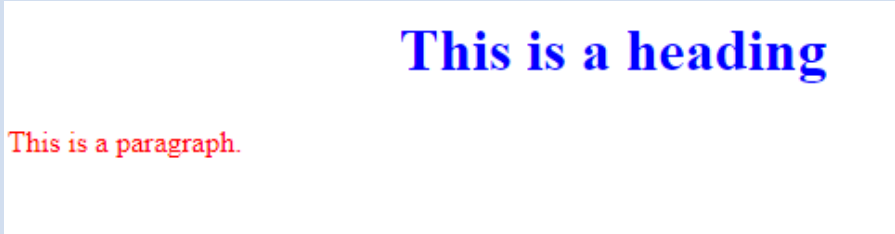
- When a browser reads a style sheet, it will format the HTML document according to the information in the style sheet.
- There are three ways to insert a style sheet.
 - Inline style
 - Internal style sheet
 - External style sheet

Inline style

- An inline style is usually used to apply a unique style for a single HTML element.
- The **style attribute** is used to add style to the relevant element.
- The style attribute can contain any CSS property.
- An inline style loses many of the advantages of a style sheet (by mixing content with presentation).
- Eg:

```
<h1 style="color:blue;text-align:center;">This is a heading</h1>
<p style="color:red;">This is a paragraph.</p>
```

Inline Style

HTML Markup	Formatted Text
<pre> <!DOCTYPE html> <html> <body> <h1 style="color:blue;text-align:center;">This is a heading</h1> <p style="color:red;">This is a paragraph.</p> </body> </html> </pre>	

Internal/embedded Style Sheets

- An internal style sheet may be used if one single HTML page has a unique style.
- Inline style sheets are difficult to modify. Discuss why?
- The internal style is defined inside the <style> element, inside the **head** section.
- HTML5 only requires style for the opening tag of an embedded style sheet.

Internal Style Sheet Example

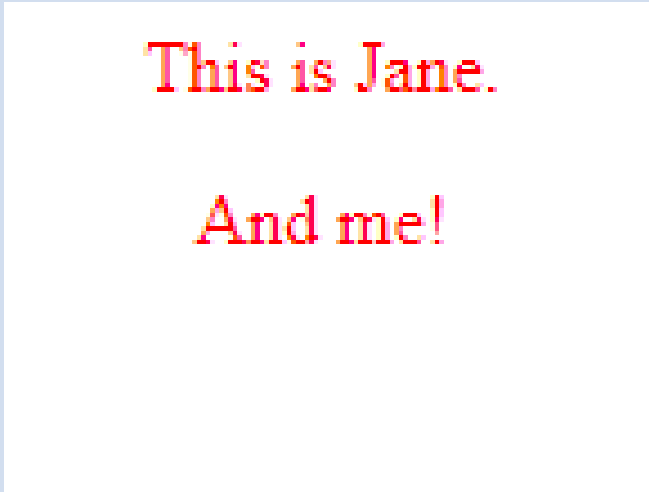
HTML Markup	Formatted Text
<pre> <!DOCTYPE html> <html> <head> <style> h1 { color: maroon; margin-left: 40px; } </style> </head> <body> <h1>This is a heading</h1> <p>This is a paragraph.</p> </body> </html> </pre>	

CSS Selectors

- CSS selectors are used to "find" (or select) the HTML elements you want to style.

Element Selector	• Selects HTML elements based on the element name
ID Selector	• Uses the id attribute of an HTML element
Class Selector	• Selects HTML elements with a specific class attribute
Universal Selector	• Selects all HTML elements on the page (*)
Grouping Selector	• Selects all the HTML elements with the same style definitions

CSS Element Selector

HTML Markup	Formatted Text
<pre> <!DOCTYPE html> <html> <head> <style> p { text-align: center; color: red; } </style> </head> <body> <p> This is Jane.</p> <p>And me!</p> </body> </html> </pre>	

CSS ID Selector

HTML Markup	Formatted Text
<pre> <!DOCTYPE html> <html> <head> <style> #para1 { text-align: center; color: red; } </style> </head> <body> <p id="para1">Hello World</p> <p>Hi.</p> </body> </html> </pre>	

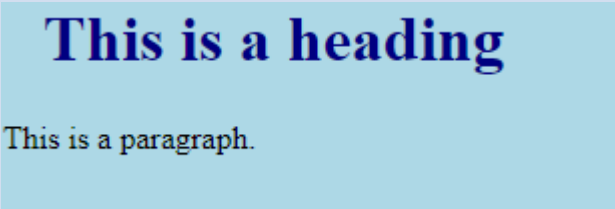
CSS Class Selector

HTML Markup	Formatted Text
<pre> <!DOCTYPE html> <html> <head> <style> .center { text-align: center; color: red; } </style> </head> <body> <h1 class="center">Red and center-aligned heading</h1> </body> </html> </pre>	

External Style Sheet

- When building a website there are several advantages to placing your CSS rules in a separate style sheet.
- All of the web pages can share the same style sheet.
- This is achieved by using the `<link>` element on each HTML page of the site to link to the same CSS document.
- Each page must include a link to the style sheet with the `<link>` tag. The `<link>` tag goes inside the head section.
- Why External CSS ? Discuss
 - Less code
 - Faster loading
 - Easy to modify
 - Uniform formatting

External Style Sheet Example

HTML Markup	Formatted Text
<pre> <!DOCTYPE html> <html> <head> <link rel="stylesheet" type="text/css" href="mystyle.css"> </head> <body> <h1>This is a heading</h1> <p>This is a paragraph.</p> </body> </html> </pre>	

External CSS Files

- An external style sheet can be written in any text editor, and must be saved with a .css extension.
- The external .css file should not contain any HTML tags.
- Here is how the "mystyle.css" file looks like:

```
body {
    background-color: lightblue;
}

h1 {
    color: navy;
    margin-left: 20px;
}
```

CSS Pseudo Classes

- There are three pseudo-classes that allow you to change the appearance of elements when a user is interacting with them.
- A pseudo-class is used to define a special state of an element. Eg:
 - Style an element when a user mouses over it
 - Style visited and unvisited links differently
 - Style an element when it gets focus
- Syntax

```
selector:pseudo-class {
    property: value;
}
```

CSS Pseudo Elements

- A CSS pseudo-element is used to style specified parts of an element.

Eg:

- Style the first letter, or line, of an element
- Insert content before, or after, the content of an element

- Syntax

-

```
selector::pseudo-element {
  property: value;
}
```

CSS Pseudo Classes/Elements Example

```
/* unvisited link */
a:link {
    color: #FF0000;
}

/* visited link */
a:visited {
    color: #00FF00;
}

/* mouse over link */
a:hover {
    color: #FF00FF;
}

/* selected link */
a:active {
    color: #0000FF;
}
```

```
p::first-line {
    color: #ff0000;
    font-variant: small-caps;
}
```

```
p::first-letter {
    color: #ff0000;
    font-size: xx-large;
}
```

```
p.intro::first-letter {
    color: #ff0000;
    font-size: 200%;
}
```


Styling Tables

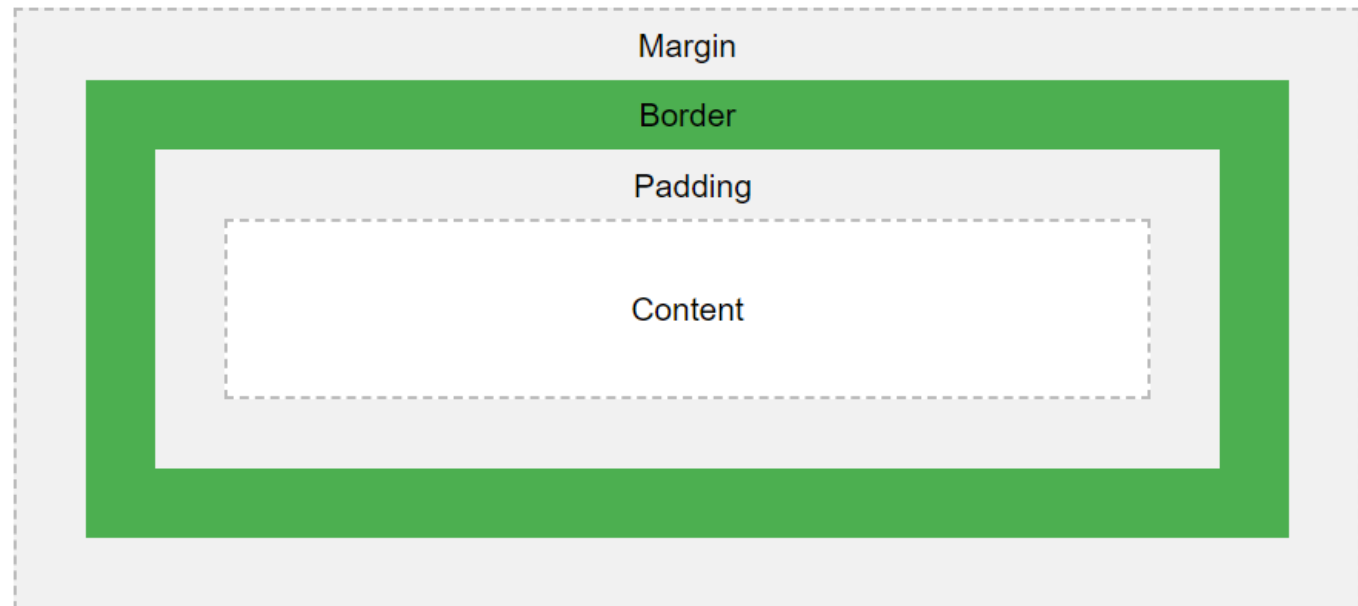
HTML Markup	CSS
<pre> <table> <caption>My First Table</caption> <tr class="b"> <th colspan="2">Col 1</th> </tr> <tr> <td rowspan="2">Cell 1</td><td>Cell 2</td> </tr> <tr class="a"> <td>Cell 4</td> </tr> </table> </pre>	<pre> <style> table,td,th{ border-collapse: collapse; border: 2px dashed blue; } td{ padding:10px;} th{text-align:left;} .a{background-color:#D0FFFF;} .b{background-color:#FFFFD0} </style> </pre>

Styling Lists

HTML Markup	CSS
<pre> AAA XXX YYY Pete Jane Rex </pre>	<pre> <style> ol{ list-style-type:square } ul{ list-style-type:lower-alpha } </style> </pre> - Item one of list - Item two of list - Item one of list - Item two of list - Item three of list

CSS Box Model

- All HTML elements can be considered as boxes.
- The CSS box model is essentially a box that wraps around every HTML element.
- It consists of:
 - Margins
 - Borders
 - Padding
 - Actual content



CSS Box Model Example

HTML Markup	Formatted Text
<pre> <!DOCTYPE html> <html> <head> <style> div { background-color: lightgrey; width: 300px; border: 15px solid green; padding: 50px; margin: 20px; } </style> </head> <body> <h2>Demonstrating the Box Model</h2> <div>Hi Hello World !!!!!</div> </body> </html> </pre>	Demonstrating the Box Model  Hi Hello World !!!!! What is this total width <div> element ? 300px (width) + 100px (left + right padding) + 30px (left + right border) + 40px (left + right margin) = 470px

How CSS Cascade?

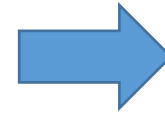
- What style will be used when there is more than one style specified for an HTML element?
- All the styles in a page will "cascade" into a new "virtual" style sheet by the following rules, where number one has the highest priority:
 - Inline style (inside an HTML element)
 - External and internal style sheets (in the head section)
- Inline style has the highest priority, and will override external and internal styles.

CSS Positioning

- CSS has the positioning schemes that allow to control the position of,
 - Layouts of pages
 - Images
- The **float** property is used for positioning and formatting.
- The float property can be used to wrap text around images.
- The float property can have following values:
 - *Float:left*
 - *Float:right*
 - *Float:none - default*
 - *Float:inherit*

Floating Images

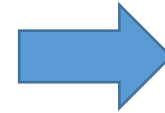
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Phasellus imperdiet, nulla et dictum interdum, nisi lorem egestas odio, vitae scelerisque enim ligula venenatis dolor. Maecenas nisl est, ultrices nec congue eget, auctor vitae massa. Fusce luctus vestibulum augue ut aliquet. Mauris ante ligula, facilisis sed ornare eu, lobortis in odio. Praesent convallis urna a lacus interdum ut hendrerit risus congue. Nunc sagittis dictum nisi, sed ullamcorper ipsum dignissim ac...



```
img {  
  float: right;  
}
```



Lorem ipsum dolor sit amet, consectetur adipiscing elit. Phasellus imperdiet, nulla et dictum interdum, nisi lorem egestas odio, vitae scelerisque enim ligula venenatis dolor. Maecenas nisl est, ultrices nec congue eget, auctor vitae massa. Fusce luctus vestibulum augue ut aliquet. Mauris ante ligula, facilisis sed ornare eu, lobortis in odio. Praesent convallis urna a lacus interdum ut hendrerit risus congue. Nunc sagittis dictum nisi, sed ullamcorper ipsum dignissim ac...

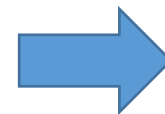


```
img {  
  float: left;  
}
```



Lorem ipsum dolor sit amet, consectetur adipiscing elit. Phasellus imperdiet, nulla et

dictum interdum, nisi lorem egestas odio, vitae scelerisque enim ligula venenatis dolor. Maecenas nisl est, ultrices nec congue eget, auctor vitae massa. Fusce luctus vestibulum augue ut aliquet. Mauris ante ligula, facilisis sed ornare eu, lobortis in odio. Praesent convallis urna a lacus interdum ut hendrerit risus congue. Nunc sagittis dictum nisi, sed ullamcorper ipsum dignissim ac...

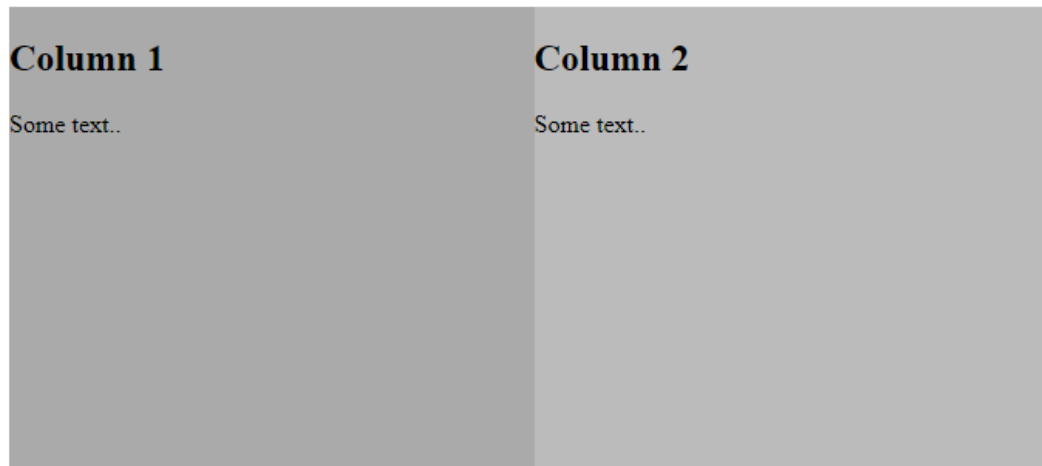


```
img {  
  float: none;  
}
```

Floating Layouts

- There are two layouts approaches available
 - Liquid Layouts with percentage
 - Fixed Layouts with pixels

Two Equal Columns



```

<head>
<style>
/* Create two equal columns that floats next to each other */
.column1{
  float:left;
  width: 50%;
  height: 300px;
}
.column2{
  float:left;
  width: 50%;
  height: 300px;
}
</style>
</head>
<body>
<h2>Two Equal Columns</h2>
<div class="column1" style="background-color:#aaa;">
  <h2>Column 1</h2>
  <p>Some text..</p>
</div>
<div class="column2" style="background-color:#bbb;">
  <h2>Column 2</h2>
  <p>Some text..</p>
</div>
</body>
</html>
  
```


CSS3

- CSS3 is the latest version of the CSS specification.
- The CSS3 specification is still under development.
- However, many of the new CSS3 properties have been implemented in modern browsers.
- CSS3 adds several new styling features and improvements to enhance the web presentation capabilities.

Review Questions

- What tag is used to link to an external style sheet?
- How do you define and use a CSS class?
- What is the difference between padding and margin?
- Write a CSS code to display following

CSS background-color example!

This is a text inside a div element.

This paragraph has its own background color.

We are still in the div element.

CSS Validators

 **Markup Validation Service**
Check the markup (HTML, XHTML, ...) of Web documents

Validate by URI

Validate by File Upload

Validate by Direct Input

Validate by URI

Validate a document online:

Address:

► [More Options](#)

Check

READING

- Duckett, J. (2011), *HTML & CSS, Design and Build Websites*, Wiley.
- Robinson, J. (2018), *Learning Web Design, A Beginner's Guide to HTML, CSS, JavaScript, And Web Graphics*, O'Reilly.
- Castro & Hyslop, B. (2011), *HTML5&CSS: Visual QuickStart Guide*, Peachpit Press.
- Online :
 - <https://www.w3schools.com/css/default.asp>